

T3 group factors: EFT _1 _after _F

Active theory: SM + $S_1 + S_2 + C_{LLS_1S_2}$. The fermion F is absent, so y_1 , y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1, m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2} \lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HH S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 – S_2 contraction

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	F	S1, S2
T3-C	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	F	S1, S2
T3-A	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	F	S1, S2
T3-D	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-E	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	F	S1, S2

Field representations

run	S_1	S_2	α
T3 – B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	–1
T3 – C	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	–1
T3 – A	$(1, 0)$	$(3, 1)$	0
T3 – D	$(3, -1)$	$(1, 0)$	–2
T3 – E	$(3, 0)$	$(3, 1)$	0

Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
g_1^3	$\frac{43}{6}$	$\frac{43}{6}$	$\frac{47}{6}$	$\frac{47}{6}$	$\frac{47}{6}$

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{1}{2}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
g_3^3	-7	-7	-7	-7	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = -3$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\beta_{m_1^2}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{S_1}^{(1)} m_1^2$	6	6	4	8	8
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	0	-12	-12
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	0	-6	0
$\lambda_{12}^{(1)} m_2^2$	4	4	6	2	6
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	4	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	4

 $\beta_{m_2^2}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{S_2}^{(1)} m_2^2$	6	6	8	4	8
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	-12	0	-12
$g_1^2 m_2^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6	0	-6
$\lambda_{12}^{(1)} m_1^2$	4	4	2	6	6
$\lambda_{S_2}^{(A)} m_2^2$	0	0	4	0	4
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	4

 $\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$ \lambda_5 ^2$	6	6	12	4	4
$g_2^2 \lambda_{HS_1}^{(1)}$	-9	-9	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	-3	-3	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	6	6	4	8	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	6	2	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	6	6
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	0	3	0
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	2	2
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	4	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	4

 $\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$ \lambda_5 ^2$	6	6	4	12	4
$g_2^2 \lambda_{HS_2}^{(1)}$	-9	-9	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	-3	-3	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	6	6	8	4	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	4	4	2	6	6

g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	6	0	6
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	3	0	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	2	0	2
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	0	4
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	4

$$\beta_{\lambda_{12}^{(1)}}$$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$ \lambda_5 ^2$	3	3	4	4	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	-9	-9	-12	-12	-24
$g_1^2 \lambda_{12}^{(1)}$	-3	-3	-6	-6	-6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	6	6	4	8	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	6	6	8	4	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	0	12
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	0	4
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	0	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	4

$$\beta_{\lambda_5}$$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	-9	-9	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	-3	-3	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	0	-4	-2
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	4	0	2
$\lambda_5 \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	0	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	-2

Additional scalar-coupling beta functions

$$\beta_{\lambda_{HS_1}^{(A)}}$$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$g_1^2 g_2^2$	-6	-6	0	-12	0

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$g_1^2 g_2^2$	6	6	12	0	12

 $\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$g_1^2 g_2^2$	−6	−6	0	0	0

 $\beta_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
g_2^4	0	0	0	0	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	−2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	−2

Direct Weinberg mixing

The active EFT1 scalar interaction produces

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1 S_2}.$$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
R_W	$\frac{4\sqrt{3}}{3}$	$-\frac{4}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$-\frac{4\sqrt{6}}{3}$

The displayed R_W values use the validated Wigner–6j reduction and inherited tensor convention. The overall tensor-level normalization/phase has not yet been independently derived from first principles.

F-threshold matching tensor samples

The component bases have representation-dependent dimensions, so the raw tensor entries are not placed in the cross-run beta-function tables.

T3 – B ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\overline{y_1}^2}{4M_F}$
C_{1156}	$-\frac{i\overline{y_1}^2}{4M_F}$
C_{11511}	$\frac{\overline{y_1 y_2}}{4M_F}$
C_{11512}	$\frac{i\overline{y_1 y_2}}{4M_F}$
C_{1165}	$-\frac{i\overline{y_1}^2}{4M_F}$
C_{1166}	$-\frac{\overline{y_1}^2}{4M_F}$

C_{11611}	$-\frac{iy_1 y_2}{4M_F}$
C_{11612}	$\frac{y_1 y_2}{4M_F}$
C_{11115}	$\frac{y_1 y_2}{4M_F}$
C_{11116}	$-\frac{iy_1 y_2}{4M_F}$

T3 – C ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\sqrt{3}y_1^2}{12M_F}$
C_{1156}	$-\frac{\sqrt{3}iy_1^2}{12M_F}$
C_{11511}	$-\frac{\sqrt{3}y_1 y_2}{12M_F}$
C_{11512}	$-\frac{\sqrt{3}iy_1 y_2}{12M_F}$
C_{1165}	$-\frac{\sqrt{3}iy_1^2}{12M_F}$
C_{1166}	$-\frac{\sqrt{3}y_1^2}{12M_F}$
C_{11611}	$\frac{\sqrt{3}iy_1 y_2}{12M_F}$
C_{11612}	$-\frac{\sqrt{3}y_1 y_2}{12M_F}$
C_{11115}	$-\frac{\sqrt{3}y_1 y_2}{12M_F}$
C_{11116}	$\frac{\sqrt{3}iy_1 y_2}{12M_F}$

T3 – A ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{1158}	$\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{11511}	$\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{11512}	$-\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{1167}	$-\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{1168}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{11611}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{1175}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{1176}	$-\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$

T3 – D ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{11512}	$\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{11611}	$-\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{11612}	$\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$
C_{11911}	$\frac{3^{\frac{3}{4}}iy_1 y_2}{12M_F}$
C_{11912}	$-\frac{3^{\frac{3}{4}}y_1 y_2}{12M_F}$

C_{111011}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111012}	$\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$
C_{111115}	$\frac{3^{\frac{3}{4}} \overline{y_1 y_2}}{12 M_F}$
C_{111116}	$-\frac{3^{\frac{3}{4}} i \overline{y_1 y_2}}{12 M_F}$

T3 – E ($\alpha = 0$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11514}	$\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11613}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11614}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11711}	$-\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11712}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11715}	$-\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11716}	$\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$
C_{11811}	$\frac{\sqrt{3} i \overline{y_1 y_2}}{12 M_F}$
C_{11812}	$-\frac{\sqrt{3} \overline{y_1 y_2}}{12 M_F}$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.